Program-5 Documentation

Title: Integrate Git with Docker for Source-Controlled Application Builds

Project Structure and Files:

The setup relies on two main files in the project directory (`~/program5$`) where the commands were executed:

- `Dockerfile`: Defines how to build a custom Docker image for Jenkins.
  - It uses `FROM jenkins/jenkins:lts` as the base image, which is the official Jenkins Long-Term Support version.
- `docker-compose.yml`: Defines the services, networks, and volumes for the multi-container Docker application.

Section	Key Configuration	Purpose
version	3.8	Specifies the Docker Compose file format version. (Note: The command output warns this attribute is obsolete in newer Docker versions.)
services: jenkins	build: .	Tells Docker Compose to build the image using the Dockerfile in the current directory.
container_name	jenkins-ci	Assigns a specific, easy-to-reference name to the running container.
ports	8080:8080, 50000:50000	Maps ports from the host machine to the container (HTTP access and agent communication, respectively).
volumes	jenkins_home:/var/jenkins_home	Persistent Storage: Maps a named Docker volume (jenkins_home) to the Jenkins configuration directory *inside* the container, ensuring data persists after container restarts/deletes.
volumes	/var/run/docker.sock:/var/run/docker.sock	"Docker in Docker" (DooD): Maps the Docker daemon socket from the host to the container. This allows the Jenkins container to execute Docker commands (e.g., building or running other containers) on the host machine.
volumes: jenkins_home:	(Empty)	Defines the named volume used for persistent storage.

Execution Steps

1. Change Directory:
   - The user navigated to the project folder: `cd program5`

```
1RV24MC061-leelanjali-p@Radha-Leelanjali:~$ cd program5
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program5$ ls
docker-compose.yml  Dockerfile
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program5$ docker compose build
WARN[0000] /home/1RV24MC061-leelanjali-p/program5/docker-compose.yml: the attribute `version` is obsolete, it will be igno
red, please remove it to avoid potential confusion
[+] Building 1.3s (7/7) FINISHED
 => [internal] load local bake definitions                                                                  0.0s
 => => reading from stdin 535B                                                                              0.0s
 => [internal] load build definition from Dockerfile                                                        0.0s
 => => transferring dockerfile: 74B                                                                         0.0s
 => [internal] load metadata for docker.io/jenkins/jenkins:lts                                              1.2s
 => [internal] load .dockerignore                                                                           0.0s
 => => transferring context: 2B                                                                             0.0s
 => CACHED [1/1] FROM docker.io/jenkins/jenkins:lts@sha256:7b1c378278279c8688efd6168c25a1c2723a6bd6f0420beb5ccefabe  0.0s
 => exporting to image                                                                                      0.0s
 => => exporting layers                                                                                     0.0s
 => => writing image sha256:e86a77a1530fa0046d88f4a655484f731d46b968e048ea5fdbd88166fbd868a0                0.0s
 => => naming to docker.io/library/program5-jenkins                                                         0.0s
 => resolving provenance for metadata file                                                                  0.0s
[+] Building 1/1
 ✔ program5-jenkins  Built                                                                                  0.0s
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program5$ docker compose up -d
WARN[0000] /home/1RV24MC061-leelanjali-p/program5/docker-compose.yml: the attribute `version` is obsolete, it will be igno
red, please remove it to avoid potential confusion
[+] Running 1/1
 ✔ Container jenkins-ci  Running                                                                            0.0s
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program5$
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program5$ docker images
REPOSITORY                TAG       IMAGE ID       CREATED          SIZE
leelanjai/program-6       latest    17ee6e32bb9e   27 minutes ago   141MB
```

2. Verify Files:

   - Confirmed the presence of the required files: `ls` (showed `docker-compose.yml` and `Dockerfile`).

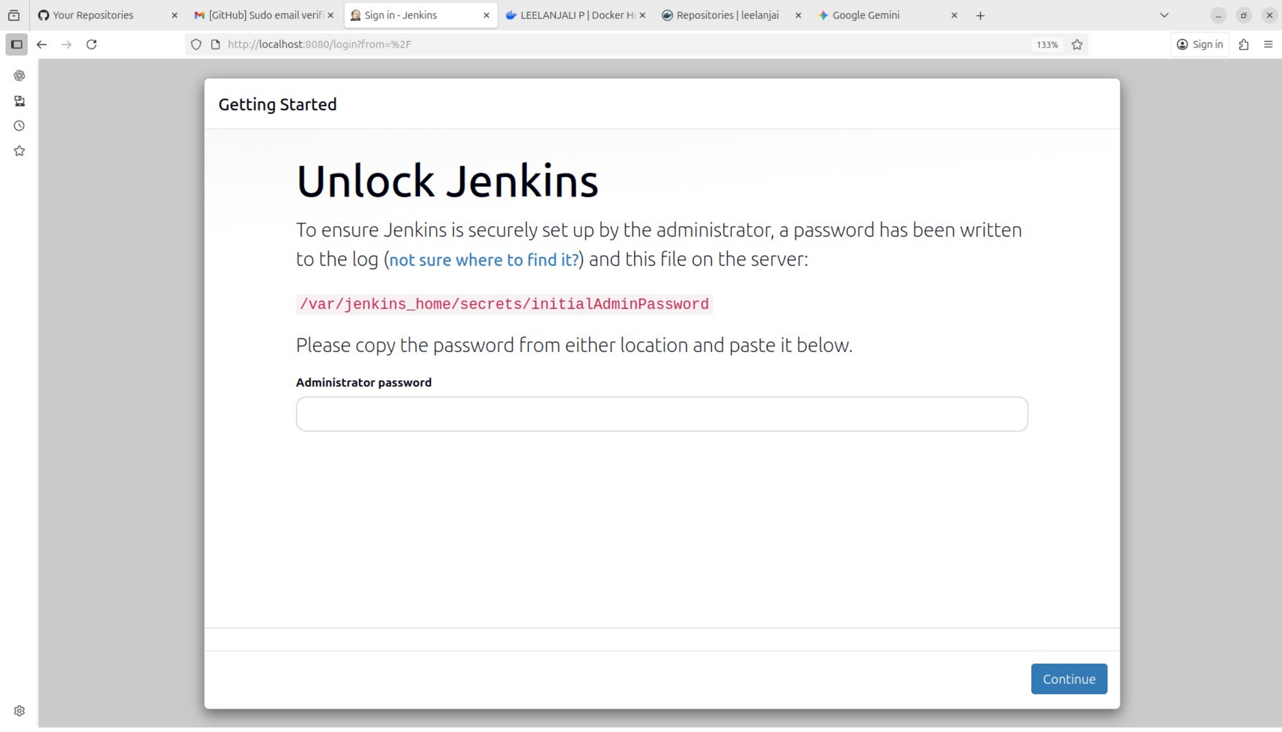
```
  GNU nano 7.2                              docker-compose.yml
version: '3.8'

services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:

                                       [ Read 15 lines ]
^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

3. Build the Image:
   - Command: `docker compose build`
   - This command reads the `Dockerfile` and builds the Jenkins image, naming it `program5-jenkins` (based on the folder name and service name).
4. Run the Container:
   - Command: `docker compose up -d`
   - This command creates and starts the `jenkins-ci` container in detached mode (`-d`). The output shows `Container jenkins-ci Running`.



5. Verify Images/Containers (Optional):
   - `docker images` was used to confirm the custom image was built and is tagged as `program5-jenkins`.